SRM Institute of Science and Technology

Faculty of Engineering and Technology

DEPARTMENT OF MATHEMATICS

21MAB204T-Probability & Queuing Theory - Assignment II

1. 4 coins were tossed simultaneously. What is the probability of getting (i) 2 heads (ii) at least 2 heads (iii)at most 2 heads.
2. A manufacturer of pins knows that 2% of his products are defective. If he sells pins in boxes of100 and guarantees that not more than 4 pins will be defective. What is the probability that a box will fail to meet the guaranteed quality?
3. If X is a Poisson variate, P(X=2)= 9P(X=4)+90P(X=6), find the mean, variance, P(X≥2) and P(X=0)
4. The mileage which a car owner gets with a certain type of tyres is a random variable having an exponential distribution with mean 40,000 kms. Find the probability that the car owner gets i) at least 20,000 kms ii) at most 30,000 kms.
5. The atoms of a radioactive element are randomly disintegrating. If every gram on an average emits 3.9α particles per second. What is the probability that during the next second the number of α particles emitted from one gram is i) at least 2 ii) at most 6
6. The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{2}$. (i).What is the probability that a repair time exceeds 2h? (ii).What is the conditional probability that a repair time takes 11h given that its duration exceeds 8h?
7. 6.In a distribution exactly normal, 7 % of the items are under 35 and 89% are under 63.Find the mean and standard deviation of the distribution.
8. In a normal distribution,31 % of the items are under 45 and 8% are over 64.Find the mean and standard deviation of the distribution
9. 8. In a town, out of 800 families with 4 children each, how many families would be expected to have i) 2 boys & 2 girls ii) atleast 1 boy iii) no girl iv) at most 2 girls v) children of both gender.
10. Six dice are thrown 729 times. How many times do you expect i) at least 3 dice ii) 2 dice to show a 5 or 6?